The core idea has a close analogue in a different domain. The **knowledge-distilled GNN for seizure detection** (EPFL, 2023) is the closest published precedent: it proposes transferring knowledge from a sophisticated seizure detector trained on a full set of EEG electrodes to learn new detectors that use significantly fewer electrodes. Both teacher and student are GNNs, since these architectures actively use connectivity information. Using as few as two channels, competitive seizure detection performance is achievable. This is intra-modal graph distillation — same signal type, fewer nodes. What your architecture needs is *cross-modal* graph distillation: EEG teacher graph → ECG/HRV/EDA student graph.

The mapping mechanism would operate as follows. During calibration, a dynamic GNN learns a subject-specific emotion-state-conditioned adjacency matrix over EEG channels — using PLV, coherence, or a learned metric. This matrix encodes which brain regions co-activate for which emotional transitions. The question then becomes: which entries in that matrix have analogues in the wearable sensor domain? ECG captures cardiac autonomic response, HRV captures parasympathetic tone, EDA captures sympathetic arousal — these are precisely the peripheral projections of the autonomic nervous system branches that the EEG functional connectivity is indexing. The transfer is not a direct adjacency copy but a *learned projection*: a small MLP trained to map EEG connectivity patterns to wearable sensor relationship weights.

The output is a personalized static graph for the wearable student — not universal (which would miss subject-specific autonomic patterns) but not re-learned from scratch on EEG each time either.